

NTP Client

Lakukan instalasi paket layanan sinkronisasi waktu

```
root@www:~# apt install systemd-timesyncd
```

Pastikan konfigurasi timezone ke Asia/Jakarta

```
sudo timedatectl set-timezone Asia/Jakarta
```

Melakukan konfigurasi Real Time Clock (RTC) untuk merefer ke UTC (Coordinated Universal Time)

```
sudo timedatectl set-local-rtc false
```

Mengaktifkan NTP Client untuk sinkronisasi waktu

```
sudo timedatectl set-ntp true
```

Menyunting file timesyncd.conf untuk mengarah ke NTP server terdekat untuk mendapatkan waktu delay terpendek. Biasanya setiap organisasi atau negara mempunyai NTP Server sendiri

```
sudo nano /etc/systemd/timesyncd.conf
```

```
#Ubahlah line 16
```

```
[Time]
```

```
NTP=0.id.pool.ntp.org
```

Restart layanan sinkronisasi waktu dan pastikan layanan berjalan dengan benar

```
sudo systemctl restart systemd-timesyncd  
sudo systemctl status systemd-timesyncd
```

Lakukan pengecekan kesesuaian tanggal system dengan perintah

```
root@www:~# timedatectl
```

Hasilnya adalah :

```
Server: 103.134.77.146 (0.id.pool.ntp.org)
Poll interval: 17min 4s (min: 32s; max 34min 8s)
Leap: normal
Version: 4
Stratum: 2
Reference: D8EF2304
Precision: 1us (-24)
Root distance: 23.680ms (max: 5s)
Offset: +1.223ms
Delay: 49.829ms
Jitter: 2.249ms
Packet count: 5
Frequency: +14.953ppm
```

Apache 2 + PHP-FM

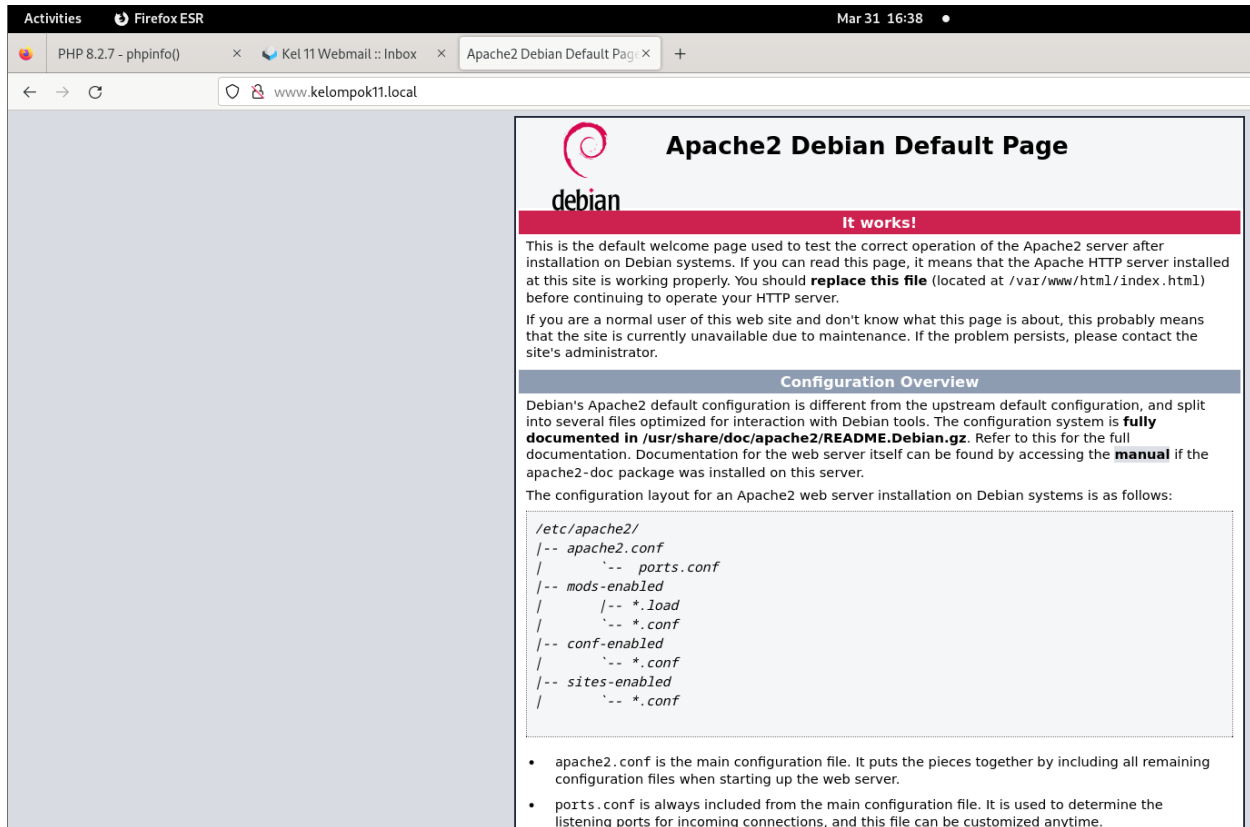
Install Apache2

```
root@www:~# apt -y install apache2
```

Mengkonfigurasi Apache2

```
root@www:~# vi /etc/apache2/conf-enabled/security.conf
# line 12 : change
ServerTokens Prod
root@www:~# vi /etc/apache2/mods-enabled/dir.conf
# add file name that it can access only with directory's name
DirectoryIndex index.html index.htm
root@www:~# vi /etc/apache2/apache2.conf
# line 70 : add to specify server name
ServerName www.kelompok11.com
root@www:~# vi /etc/apache2/sites-enabled/000-default.conf
# line 11 : change to webmaster's email
ServerAdmin webmaster@kelompok11.com
root@www:~# systemctl reload apache2
```

Melakukan test ke web browser



Install PHP 8.2

```
root@www:~# apt -y install php8.2 php8.2-mbstring php-pear  
root@www:~# php -v  
PHP 8.2.7 (cli) (built: Jun 9 2023 19:37:27) (NTS)  
Copyright (c) The PHP Group  
Zend Engine v4.2.7, Copyright (c) Zend Technologies  
with Zend OPcache v8.2.7, Copyright (c), by Zend Technologies  
  
# verify installation to create a test script  
root@www:~# echo '<?php echo `php -i`."\\n"; ?>' > php_test.php  
root@www:~# php php_test.php | head  
phpinfo()  
PHP Version => 8.2.7  
  
System => Linux dlp.srv.world 6.1.0-9-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.27-1  
(2023-05-08) x86_64  
Build Date => Jun 9 2023 19:37:27  
Build System => Linux  
Server API => Command Line Interface  
Virtual Directory Support => disabled  
Configuration File (php.ini) Path => /etc/php/8.2/cli  
Loaded Configuration File => /etc/php/8.2/cli/php.ini
```

Install PHP-FM

```
root@www:~# apt -y install php-fpm
```

Mengkonfigurasi PHP-FM pada file konfigurasi Apache

```
root@www:~# vi /etc/apache2/sites-available/default-ssl.conf
# add into <VirtualHost> - </VirtualHost>
    <FilesMatch \.php$>
        SetHandler "proxy:unix:/var/run/php/php8.2-fpm.sock|fcgi://localhost/"
    </FilesMatch>
</VirtualHost>

root@www:~# a2enmod proxy_fcgi setenvif
Considering dependency proxy for proxy_fcgi:
Enabling module proxy.
Enabling module proxy_fcgi.
Module setenvif already enabled
To activate the new configuration, you need to run:
    systemctl restart apache2

root@www:~# a2enconf php8.2-fpm
Enabling conf php8.1-fpm.
To activate the new configuration, you need to run:
    systemctl reload apache2

root@www:~# systemctl restart php8.2-fpm apache2
```

Melakukan test validasi terhadap PHP-FM dengan membuat file info.php di root document

```
root@www:~# echo '<?php phpinfo(); ?>' > /var/www/html/info.php
```

Lalu lakukan test di browser

PHP 8.2.7 - phpinfo() Kel 11 Webmail :: Inbox PHP 8.2.7 - phpinfo() ← → ↻ www.kelompok11.local/info.php	
PHP Version 8.2.7 	
System	Linux debian12-01 6.1.0-18-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.76-1 (2024-02-01) x86_64
Build Date	Jun 9 2023 19:37:27
Build System	Linux
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.2/fpm
Loaded Configuration File	/etc/php/8.2/fpm/php.ini
Scan this dir for additional .ini files	/etc/php/8.2/fpm/conf.d
Additional .ini files parsed	/etc/php/8.2/fpm/conf.d/10-mysqld.ini, /etc/php/8.2/fpm/conf.d/10-opcache.ini, /etc/php/8.2/fpm/conf.d/10-pdo.ini, /etc/php/8.2/fpm/conf.d/15-xml.ini, /etc/php/8.2/fpm/conf.d/20-bz2.ini, /etc/php/8.2/fpm/conf.d/20-calendar.ini, /etc/php/8.2/fpm/conf.d/20-ctype.ini, /etc/php/8.2/fpm/conf.d/20-curl.ini, /etc/php/8.2/fpm/conf.d/20-dom.ini, /etc/php/8.2/fpm/conf.d/20-enchant.ini, /etc/php/8.2/fpm/conf.d/20-exif.ini, /etc/php/8.2/fpm/conf.d/20-ffi.ini, /etc/php/8.2/fpm/conf.d/20-fileinfo.ini, /etc/php/8.2/fpm/conf.d/20-ftp.ini, /etc/php/8.2/fpm/conf.d/20-gd.ini, /etc/php/8.2/fpm/conf.d/20-gettext.ini, /etc/php/8.2/fpm/conf.d/20-iconv.ini, /etc/php/8.2/fpm/conf.d/20-intl.ini, /etc/php/8.2/fpm/conf.d/20-mbstring.ini, /etc/php/8.2/fpm/conf.d/20-mcrypt.ini, /etc/php/8.2/fpm/conf.d/20-mysql.ini, /etc/php/8.2/fpm/conf.d/20-pdo_mysql.ini, /etc/php/8.2/fpm/conf.d/20-pear.ini, /etc/php/8.2/fpm/conf.d/20-posix.ini, /etc/php/8.2/fpm/conf.d/20-readline.ini, /etc/php/8.2/fpm/conf.d/20-shmop.ini, /etc/php/8.2/fpm/conf.d/20-simplexml.ini, /etc/php/8.2/fpm/conf.d/20-sockets.ini, /etc/php/8.2/fpm/conf.d/20-sysvmsg.ini, /etc/php/8.2/fpm/conf.d/20-sysvsem.ini, /etc/php/8.2/fpm/conf.d/20-sysvshm.ini, /etc/php/8.2/fpm/conf.d/20-tokenizer.ini, /etc/php/8.2/fpm/conf.d/20-xmlreader.ini, /etc/php/8.2/fpm/conf.d/20-xmlwriter.ini, /etc/php/8.2/fpm/conf.d/20-xsl.ini, /etc/php/8.2/fpm/conf.d/20-zip.ini
PHP API	20220829
PHP Extension	20220829
Zend Extension	420220829
Zend Extension Build	API420220829,NTS
PHP Extension Build	API20220829,NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
Zend Max Execution Timers	disabled

Database System : MariaDB

Lakukan nstalasi Maria DB 10.11

```

root@www:~# apt -y install mariadb-server
root@www:~# vi /etc/mysql/mariadb.conf.d/50-server.cnf
# line 95 : confirm default charset
# if use 4 bytes UTF-8, specify [utf8mb4]
character-set-server = utf8mb4
collation-server    = utf8mb4_general_ci

root@www:~# systemctl restart mariadb

```

Inisial Konfigurasi dan testing database MariaDB Server

```

root@www:~# mysql_secure_installation

```

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current password for the root user. If you've just installed MariaDB, and haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

```
# Switch to [unix_socket] authentication or not
# [unix_socket] auth is enabled for root user by default even if you select [No]
Switch to unix_socket authentication [Y/n] n
... skipping.
```

You already have your root account protected, so you can safely answer 'n'.

```
# set MariaDB root password or not
# [unix_socket] authentication is enabled by default, but
# if you set root password, it's also possible to login with password authentication.
# if not set root password, only OS root user can login as MariaDB root user
Change the root password? [Y/n] n
... skipping.
```

By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

```
# remove anonymous users
Remove anonymous users? [Y/n] y
... Success!
```

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

```
# disallow root login remotely
Disallow root login remotely? [Y/n] y
... Success!
```

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

```
# remove test database
```

```
Remove test database and access to it? [Y/n] y
```

```
- Dropping test database...
```

```
... Success!
```

```
- Removing privileges on test database...
```

```
... Success!
```

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

```
# reload privilege tables
```

```
Reload privilege tables now? [Y/n] y
```

```
... Success!
```

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

Thanks for using MariaDB!

```
# connect to MariaDB
```

```
root@www:~# mysql
```

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.
```

```
Your MariaDB connection id is 37
```

```
Server version: 10.11.3-MariaDB-1 Debian 12
```

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
# [Unix_Socket] authentication is enabled by default
```

```
MariaDB [(none)]> show grants for root@localhost;
```

```
+-----+
+-----+
| Grants for root@localhost                                     |
+-----+
+-----+
| GRANT ALL PRIVILEGES ON *.* TO `root`@`localhost` IDENTIFIED VIA mysql_native_password
USING 'invalid' OR unix_socket WITH GRANT OPTION |
| GRANT PROXY ON '@' TO 'root'@'localhost' WITH GRANT OPTION
|
+-----+
+-----+
2 rows in set (0.000 sec)
```

```
# show user list
```

```
MariaDB [(none)]> select user,host,password from mysql.user;
```

```

+-----+-----+-----+
| User   | Host   | Password |
+-----+-----+-----+
| mariadb.sys | localhost |      |
| root     | localhost | invalid |
| mysql    | localhost | invalid |
+-----+-----+-----+
3 rows in set (0.001 sec)

```

```

# show database list
MariaDB [(none)]> show databases;

```

```

+-----+
| Database      |
+-----+
| information_schema |
| mysql          |
| performance_schema |
| sys            |
+-----+
4 rows in set (0.000 sec)

```

```

# create test database
MariaDB [(none)]> create database test_database;
Query OK, 1 row affected (0.000 sec)

```

```

# create test table on test database
MariaDB [(none)]> create table test_database.test_table (id int, name varchar(50), address
varchar(50), primary key (id));
Query OK, 0 rows affected (0.108 sec)

```

```

# insert data to test table
MariaDB [(none)]> insert into test_database.test_table(id, name, address) values("001", "Debian",
"Hiroshima");
Query OK, 1 row affected (0.036 sec)

```

```

# show test table
MariaDB [(none)]> select * from test_database.test_table;
+---+-----+-----+
| id | name  | address |
+---+-----+-----+
| 1  | Debian | Hiroshima |
+---+-----+-----+
1 row in set (0.000 sec)

```

```

# delete test database
MariaDB [(none)]> drop database test_database;
Query OK, 1 row affected (0.111 sec)

```



```
MariaDB [(none)]> exit
Bye
```

Email System

POSTFIX : SMTP Server (TCP 25)

```
root@mail:~# apt -y install postfix sasl2-bin
# on this example, proceed to select [No Configuration]
# because configure all manually
+-----+ Postfix Configuration +-----+
| General type of mail configuration: |
|                                     |
|   No configuration                 |
|   Internet Site                   |
|   Internet with smarthost         |
|   Satellite system               |
|   Local only                     |
|                                     |
|   <Ok>      <Cancel>              |
|                                     |
+-----+

root@mail:~# cp /usr/share/postfix/main.cf.dist /etc/postfix/main.cf
root@mail:~# vi /etc/postfix/main.cf
# line 82 : uncomment
mail_owner = postfix

# line 98 : uncomment and specify hostname
myhostname = mail.kelompok11.local

# line 106 : uncomment and specify domainname
mydomain = kelompok11.local

# line 127 : uncomment
myorigin = $mydomain

# line 141 : uncomment
inet_interfaces = all

# line 189 : uncomment
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
```

```
# line 232 : uncomment
local_recipient_maps = unix:passwd.byname $alias_maps

# line 277 : uncomment
mynetworks_style = subnet

# line 294 : add your local network
mynetworks = 127.0.0.0/8, 10.0.0.0/24, 192.168.0.0/16

# line 416 : uncomment
alias_maps = hash:/etc/aliases

# line 427 : uncomment
alias_database = hash:/etc/aliases

# line 449 : uncomment
home_mailbox = Maildir/

# line 585: comment out and add
#smtpd_banner = $myhostname ESMTP $mail_name (Debian/GNU)
smtpd_banner = $myhostname ESMTP

# line 659 : add
sendmail_path = /usr/sbin/postfix

# line 664 : add
newaliases_path = /usr/bin/newaliases

# line 669 : add
mailq_path = /usr/bin/mailq

# line 675 : add
setgid_group = postdrop

# line 679 : comment out
#html_directory =

# line 683 : comment out
#manpage_directory =

# line 688 : comment out
#sample_directory =

# line 692 : comment out
#readme_directory =

# line 692 : if also listen IPv6, change to [all]
inet_protocols = ipv4
```

```
# add follows to the end
# disable SMTP VRFY command
disable_vrfy_command = yes

# require HELO command to sender hosts
smtpd_helo_required = yes

# limit an email size
# example below means 10M bytes limit
message_size_limit = 10240000

# SMTP-Auth settings
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
smtpd_recipient_restrictions = permit_mynetworks, permit_auth_destination,
permit_sasl_authenticated, reject

root@mail:~# newaliases
root@mail:~# systemctl restart postfix
```

Menambahkan konfigurasi anti spam

```
root@mail:~# vi /etc/postfix/main.cf
# add to the end
# reject unknown clients that forward lookup and reverse lookup of their hostnames on DNS do
not match
smtpd_client_restrictions = permit_mynetworks, reject_unknown_client_hostname, permit

# rejects senders that domain name set in FROM are not registered in DNS or
# not registered with FQDN
smtpd_sender_restrictions = permit_mynetworks, reject_unknown_sender_domain,
reject_non_fqdn_sender

# reject hosts that domain name set in FROM are not registered in DNS or
# not registered with FQDN when your SMTP server receives HELO command
smtpd_helo_restrictions = permit_mynetworks, reject_unknown_hostname,
reject_non_fqdn_hostname, reject_invalid_hostname, permit

root@mail:~# systemctl restart postfix
```

DOVECOT : IMAP4 (TCP 143) and POP3 (TCP110) Server

Lakukan instalasi Dovecot Server

```
root@mail:~# apt -y install dovecot-core dovecot-pop3d dovecot-imapd
root@mail:~# vi /etc/dovecot/dovecot.conf
# line 30 : uncomment
listen = *, ::
root@mail:~# vi /etc/dovecot/conf.d/10-auth.conf
# line 10 : uncomment and change (allow plain text auth)
disable_plaintext_auth = no
# line 100 : add
auth_mechanisms = plain login
root@mail:~# vi /etc/dovecot/conf.d/10-mail.conf
# line 30 : change to Maildir
mail_location = maildir:~/Maildir
root@mail:~# vi /etc/dovecot/conf.d/10-master.conf
# line 107-109 : uncomment and add
# Postfix smtp-auth
unix_listener /var/spool/postfix/private/auth {
    mode = 0666
    user = postfix
    group = postfix
}

root@mail:~# systemctl restart dovecot
```

FINAL CHECK untuk semua SERVICES :

```
# netstat -a| grep LISTEN
```

Akan terlihat hasilnya seperti dibawah, dengan status Server (LISTEN) : **MariaDB(MySQL), IMAP, POP3, DNS(domain), IMAPS, POP3S, SSH, Postfix (SMTP)**

tcp	0	0 localhost:mysql	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:imap2	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:pop3	0.0.0.0:*	LISTEN
tcp	0	0 localhost:domain	0.0.0.0:*	LISTEN
tcp	0	0 localhost:domain	0.0.0.0:*	LISTEN
tcp	0	0 localhost:domain	0.0.0.0:*	LISTEN
tcp	0	0 localhost:domain	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:smtp	0.0.0.0:*	LISTEN
tcp	0	0 ns.kelompok11.lo:domain	0.0.0.0:*	LISTEN
tcp	0	0 ns.kelompok11.lo:domain	0.0.0.0:*	LISTEN
tcp	0	0 ns.kelompok11.lo:domain	0.0.0.0:*	LISTEN
tcp	0	0 ns.kelompok11.lo:domain	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:ssh	0.0.0.0:*	LISTEN
tcp	0	0 localhost:ipp	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:imaps	0.0.0.0:*	LISTEN
tcp	0	0 0.0.0.0:pop3s	0.0.0.0:*	LISTEN
tcp	0	0 localhost:953	0.0.0.0:*	LISTEN
tcp	0	0 localhost:953	0.0.0.0:*	LISTEN
tcp	0	0 localhost:953	0.0.0.0:*	LISTEN
tcp	0	0 localhost:953	0.0.0.0:*	LISTEN
tcp6	0	0 localhost:domain	:::*	LISTEN
tcp6	0	0 localhost:domain	:::*	LISTEN
tcp6	0	0 localhost:domain	:::*	LISTEN
tcp6	0	0 localhost:domain	:::*	LISTEN
tcp6	0	0 localhost:ipp	:::*	LISTEN
tcp6	0	0 debian12-01:domain	:::*	LISTEN
tcp6	0	0 debian12-01:domain	:::*	LISTEN
tcp6	0	0 debian12-01:domain	:::*	LISTEN
tcp6	0	0 debian12-01:domain	:::*	LISTEN
tcp6	0	0 [::]:imap2	:::*	LISTEN
tcp6	0	0 [::]:http	:::*	LISTEN
tcp6	0	0 [::]:pop3	:::*	LISTEN
tcp6	0	0 [::]:ssh	:::*	LISTEN
tcp6	0	0 [::]:imaps	:::*	LISTEN
tcp6	0	0 [::]:pop3s	:::*	LISTEN

Melakukan Cek terhadap Layanan Posfix

```
#telnet mail.kelompok11.local 25
```

Hasilnya :

```
Trying 192.168.11.10...
Connected to mail.kelompok11.local.
Escape character is '^]'.
220 mail.kelompok11.local ESMTD Postfix (Debian/GNU)
ehlo mail.kelompok11.local
250-mail.kelompok11.local
250-PIPELINING
250-SIZE 10240000
250-ETRN
250-AUTH PLAIN LOGIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
quit
221 2.0.0 Bye
Connection closed by foreign host
```